
Term Loan Interest Rate Pricing Model

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Candidate

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Abstract

This report presents a quantitative framework for pricing the interest rate of term loans. Term loan pricing is decomposed into a risk-free component, proxied by the US Treasury yield of the same maturity, and a credit spread that compensates the lender for default risk, recovery uncertainty, and illiquidity. We survey the academic literature on defaultable bond and loan pricing, propose a two-stage machine-learning model, discuss its applicability to private and unlisted borrowers, extend the framework to one-month resale price forecasting, and demonstrate how to compute 95% confidence intervals around both the static price and the one-month forecast. Code and data used in this study are available at:

<https://github.com/PLACEHOLDER/loan-pricing>

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1 Introduction

A **term loan** is a debt instrument in which the borrower pays a fixed periodic coupon and repays the principal in full at maturity. The all-in interest rate charged by the bank consists of two additive components:

$$r_{\text{loan}} = r_f(T) + s, \quad (1)$$

where $r_f(T)$ is the yield of the US Treasury bond with the same maturity T , and $s > 0$ is the *credit spread*, which compensates the lender for (i) the probability that the borrower defaults (PD), (ii) the loss given default (LGD), and (iii) the illiquidity of the loan instrument.

The objective of this project is to build a data-driven model that recommends the appropriate credit spread s for a new loan, based on borrower characteristics, loan terms, and prevailing market conditions.

2 Model Choice and Literature Survey

2.1 Overview of the Pricing Problem

Under the risk-neutral measure, the credit spread of a defaultable zero-coupon bond (and by extension a term loan) can be expressed as (Duffie and Singleton, 1999):

$$s \approx \lambda \cdot LGD, \quad (2)$$

where λ is the risk-neutral default intensity and $LGD = 1 - R$, with R the recovery rate. This fundamental relationship motivates the two-stage modelling strategy adopted in Section 2.5: first estimate λ (equivalently, the probability of default), then map (λ, LGD) to a market spread.

2.2 Structural Models

Structural models treat default as an endogenous economic event triggered when the market value of the firm's assets falls below a threshold.

Table 1: Key structural models for credit spread pricing.

Model	Year	Key Contribution
Merton (1974)	1974	Equity is a call option on firm assets; default at maturity when $V_T < D$.
Black and Cox (1976)	1976	First-passage time; default can occur before maturity.
Longstaff and Schwartz (1995)	1995	Stochastic interest rates with mean reversion.
Leland (1994)	1994	Endogenous leverage and default trigger; tax shields.
KMV / Moody's Analytics	1990s	Distance-to-Default $\rightarrow$ Expected Default Frequency (EDF).

The Merton (1974) model defines the distance to default as

$$DD = \frac{\ln(V_A/D) + (\mu - \frac{1}{2}\sigma_A^2)T}{\sigma_A\sqrt{T}}, \quad (3)$$

with V_A the asset value, D the face value of debt, μ the drift, and σ_A the asset volatility. **Limitation:** these models require observable equity prices and implied volatilities, making them inapplicable to private borrowers.

2.3 Reduced-Form (Intensity) Models

Reduced-form models treat the default event as an exogenous Poisson process governed by a hazard rate (intensity) λ_t . The price of a defaultable zero-coupon bond under the risk-neutral measure is

$$P(t, T) = \mathbb{E}^Q \left[\exp \left(- \int_t^T (r_s + \lambda_s \cdot \text{LGD}) ds \right) \right]. \quad (4)$$

Table 2: Key reduced-form models.

Model	Year	Key Contribution
Jarrow and Turnbull (1995)	1995	Default = Poisson process; constant intensity.
Duffie and Singleton (1999)	1999	Stochastic λ ; spread $\approx \lambda \cdot \text{LGD}$ in closed form.
Jarrow et al. (1997)	1997	Rating-based Markov chain transition model.
Lando (1998)	1998	Cox (doubly stochastic) process; λ driven by state variables.

2.4 Machine Learning and Empirical Models

Recent empirical work has demonstrated that accounting-based and hybrid ML models achieve competitive or superior predictive accuracy compared to structural and reduced-form models, especially for cross-sectional pricing (Altman, 1968; Mai et al., 2019).

- **Altman Z-Score (Altman, 1968):** A linear discriminant function of five financial ratios, serving as the canonical accounting-based default predictor.
- **Logistic Regression:** Regress default indicator on financial ratios; output is a calibrated PD.
- **Gradient Boosting (XGBoost / LightGBM):** State-of-the-art for tabular financial data; handles non-linearities and interactions automatically (Chen and Guestrin, 2016).
- **Deep Learning (Mai et al., 2019):** Multi-layer perceptrons and LSTMs applied to sequential financial statements.
- **Conformal Prediction:** Model-agnostic framework for constructing coverage-guaranteed prediction intervals (Angelopoulos and Bates, 2023).

2.5 Chosen Model Architecture

We adopt a **two-stage gradient-boosted tree model**:

1. **Stage 1 — PD Estimation:** An XGBoost classifier trained on borrower financial ratios outputs the probability of default $\widehat{\text{PD}}$.
2. **Stage 2 — Spread Regression:** An XGBoost regressor maps $(\widehat{\text{PD}}, \text{loan features, macro variables})$ to the credit spread $\hat{s}$.

The all-in rate is then computed via Equation (1). This architecture works for both public and private borrowers (see Section 5) and readily admits uncertainty quantification via quantile regression (see Section 7).

3 Data Sources and Features

3.1 Data Sources

Table 3: Data sources used in model training.

Source	Content	Access
SEC EDGAR XBRL API	Standardised financial statements (10-K/10-Q)	Free
FRED (St. Louis Fed)	Treasury yields, macro indices, IG/HY OAS	Free
WRDS DealScan	Historical syndicated loan spreads	Academic
SBA 7(a) Loan Data	Loan outcomes for SME borrowers	Free
ICE BofA Indices (FRED)	Investment-grade / high-yield OAS	Free
Compustat	Standardised financial ratios	WRDS/Paid

3.2 Feature Set

Table 4: Model features grouped by category.

Borrower Financials	Loan Characteristics	Market / Macro
Debt / EBITDA	Maturity (years)	Treasury yield (same maturity)
Interest Coverage Ratio	Senior / subordinated	VIX
Net Debt / Equity	Secured / unsecured	IG OAS (BAMLC0A0CM)
FCF / Total Debt	Loan size (\$M)	HY OAS (BAMLH0A0HYM2)
Revenue Growth (YoY)	Covenant package	GDP growth (YoY)
EBITDA Margin	Industry (NAICS)	Yield curve slope (10Y-2Y)
Current Ratio	Borrower credit rating	Unemployment rate

3.3 Dataset Summary

[PLACEHOLDER] Insert dataset summary statistics table here. Include: number of observations, date range, mean/median credit spread, default rate, distribution of loan maturities, and industry breakdown. Example output from `notebooks/01_edu.ipynb`.

[PLACEHOLDER] Insert credit spread distribution histogram here (output of notebooks/01_eda.ipynb, Figure 1).

4 Model Training and Evaluation

4.1 Stage 1: Probability of Default Model

The first stage trains an XGBoost classifier on borrower financial ratios to predict a binary default indicator ($y = 1$ if the borrower defaulted within the loan term).

```

1 # <<< PLACEHOLDER: paste final Stage 1 training code here >>>
2 import xgboost as xgb
3 from sklearn.model_selection import StratifiedKFold
4 from sklearn.metrics import roc_auc_score
5
6 pd_model = xgb.XGBClassifier(
7     n_estimators=...,          # PLACEHOLDER
8     max_depth=...,            # PLACEHOLDER
9     learning_rate=...,        # PLACEHOLDER
10    scale_pos_weight=...,      # PLACEHOLDER (handles class imbalance)
11    eval_metric='auc',
12    random_state=42,
13 )
14 pd_model.fit(X_train_pd, y_train_pd,
15             eval_set=[(X_val_pd, y_val_pd)],
16             early_stopping_rounds=50)

```

Listing 1: Stage 1 – XGBoost PD classifier (excerpt from `src/models/pd_model.py`).

[PLACEHOLDER] Insert Stage 1 evaluation metrics: ROC-AUC (train/val/test), PR-AUC, Brier score, calibration curve. Output of notebooks/02_pd_model.ipynb.

[PLACEHOLDER] Insert SHAP feature importance bar chart for the PD model (top 15 features). Output of notebooks/02_pd_model.ipynb, Figure 2.

4.2 Stage 2: Credit Spread Regression

The second stage trains an XGBoost regressor to predict the observed credit spread s from the estimated PD and additional features.

```

1 # <<< PLACEHOLDER: paste final Stage 2 training code here >>>
2 spread_model = xgb.XGBRegressor(

```

```

3     n_estimators=...,          # PLACEHOLDER
4     max_depth=...,            # PLACEHOLDER
5     learning_rate=...,        # PLACEHOLDER
6     objective='reg:squarederror',
7     random_state=42,
8 )
9 spread_model.fit(X_train_spread, y_train_spread,
10                  eval_set=[(X_val_spread, y_val_spread)],
11                  early_stopping_rounds=50)

```

Listing 2: Stage 2 – XGBoost spread regressor (excerpt from `src/models/spread_model.py`).

[PLACEHOLDER] Insert Stage 2 evaluation metrics: RMSE, MAE, R^2 , mean absolute percentage error (MAPE) on held-out test set. Scatter plot of predicted vs. actual spread. Output of `notebooks/03_spread_model.ipynb`.

[PLACEHOLDER] Insert SHAP feature importance bar chart and summary beeswarm plot for the spread model. Output of `notebooks/03_spread_model.ipynb`, Figures 3–4.

4.3 Hyperparameter Tuning

[PLACEHOLDER] Describe the hyperparameter search procedure. Include: search method (e.g., Optuna TPE sampler), parameter grid, number of trials, cross-validation strategy, and the final selected hyperparameters for both stages. Summarise in a table. Output of `notebooks/04_hyperparameter_tuning.ipynb`.

5 Application to Private / Unlisted Borrowers

5.1 The Challenge

Public-market models (Merton, KMV) are inapplicable to private borrowers because:

- No observable equity price or implied volatility.
- No market-traded CDS or bonds from which to extract spreads.
- Limited or unaudited financial disclosure.

5.2 Approach 1: Altman Z"-Score

Altman and Saunders (1997) adapted the original Z-Score for private firms by replacing market capitalisation with book equity:

$$Z'' = 6.56X_1 + 3.26X_2 + 6.72X_3 + 1.05X_4, \quad (5)$$

where $X_1 = \frac{\text{Working Capital}}{\text{Total Assets}}$, $X_2 = \frac{\text{Retained Earnings}}{\text{Total Assets}}$, $X_3 = \frac{\text{EBIT}}{\text{Total Assets}}$, $X_4 = \frac{\text{Book Equity}}{\text{Total Liabilities}}$.

The Z'' score is mapped to an internal credit rating and, subsequently, to a PD and spread.

5.3 Approach 2: Transfer Learning from Public Firms

Our two-stage model uses *only accounting-based features* — no market prices. It is therefore directly applicable to private borrowers: supply the financial statements, and the model returns $\widehat{\text{PD}}$ and $\hat{s}$.

5.4 Approach 3: Internal Ratings-Based (IRB) Scorecard

Consistent with the Basel III IRB approach, we build a scorecard that assigns the borrower to an internal rating bucket $r \in \{1, \dots, 10\}$:

$$\hat{s}(r) = \bar{s}_r + \beta_1 \cdot \text{Maturity} + \beta_2 \cdot \text{Collateral} + \varepsilon, \quad (6)$$

where $\bar{s}_r$ is the historically observed median spread for rating bucket r . A **private company illiquidity premium** of approximately 100–200 bps is added on top to compensate for the reduced secondary market liquidity of private loans.

[PLACEHOLDER] Insert worked example: apply the model to a synthetic private borrower with given financial ratios and loan terms. Show the pipeline output: Z"-score, internal rating, estimated PD, estimated spread, and all-in interest rate. Output of `notebooks/05_private_borrower_example.ipynb`.

6 Forecasting the Resale Price After One Month

6.1 Loan Mark-to-Market Pricing

The mark-to-market price of a fixed-rate term loan at time t is:

$$P(t) = \sum_{k=1}^n \frac{C}{(1 + r_f(t) + s(t))^{t_k}} + \frac{F}{(1 + r_f(t) + s(t))^T}, \quad (7)$$

where C is the periodic coupon, F the face value, and T the maturity. A change in price over one month $[t, t + \Delta t]$ is therefore driven by (i) changes in the Treasury yield Δr_f and (ii) changes in the credit spread Δs .

6.2 Spread Dynamics: Ornstein-Uhlenbeck Model

We model the credit spread as a mean-reverting Ornstein-Uhlenbeck (OU) process:

$$ds_t = \kappa(\theta - s_t) dt + \sigma dW_t, \quad (8)$$

where $\kappa > 0$ is the mean-reversion speed, θ is the long-run mean spread, σ is the spread volatility, and W_t is a standard Brownian motion. The one-month conditional distribution of $s_{t+\Delta t}$ given s_t is Gaussian:

$$s_{t+\Delta t} \mid s_t \sim \mathcal{N}\left(\theta + (s_t - \theta)e^{-\kappa\Delta t}, \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa\Delta t})\right). \quad (9)$$

The Treasury yield dynamics are modelled separately via a Vasicek process, calibrated to the current yield curve.

6.3 Monte Carlo Forecasting Procedure

1. Estimate OU parameters (κ, θ, σ) from historical spread time series via MLE.
2. Estimate Vasicek parameters for the Treasury yield.
3. Simulate $N = 10,000$ paths of $(r_f(t + \Delta t), s(t + \Delta t))$ over $\Delta t = 1/12$ year.
4. For each path, compute the loan price $P(t + \Delta t)$ using Equation (5).
5. Report the mean forecast $\hat{P}$ and the empirical 2.5th/97.5th percentiles.

[PLACEHOLDER] Insert Monte Carlo forecast output: histogram of simulated loan prices at $t + 1$ month, with mean, 2.5th, and 97.5th percentiles marked. Output of `notebooks/06_monte_carlo_forecast.ipynb`, Figure 5.

[PLACEHOLDER] Insert table of OU calibration results: estimated $\hat{\kappa}$, $\hat{\theta}$, $\hat{\sigma}$, along

with standard errors and log-likelihood. Output of `notebooks/06_monte_carlo_forecast.ipynb`.

7 Uncertainty Quantification: 95% Confidence Intervals

7.1 Static Pricing Interval (Current Fair Spread)

We implement **quantile gradient boosting** to produce direct 95% prediction intervals for the credit spread without distributional assumptions:

$$\hat{s}_\alpha(x) = \arg \min_{f \in \mathcal{F}} \mathbb{E}[\rho_\alpha(s - f(x))], \quad \rho_\alpha(u) = u(\alpha - \mathbf{1}_{u < 0}), \quad (10)$$

where ρ_α is the pinball (quantile) loss. We train three models: $\alpha = 0.025$, $\alpha = 0.5$ (point estimate), and $\alpha = 0.975$.

```

1  # <<< PLACEHOLDER: paste final quantile model code here >>>
2  model_low = xgb.XGBRegressor(
3      objective='reg:quantileerror',
4      quantile_alpha=0.025,
5      ... # PLACEHOLDER: hyperparameters
6  )
7  model_high = xgb.XGBRegressor(
8      objective='reg:quantileerror',
9      quantile_alpha=0.975,
10     ... # PLACEHOLDER: hyperparameters
11 )
12 model_low.fit(X_train, y_train)
13 model_high.fit(X_train, y_train)
14
15 ci_lower = model_low.predict(X_new)
16 ci_upper = model_high.predict(X_new)

```

Listing 3: Quantile regression with XGBoost (excerpt from `src/models/quantile_model.py`).

We additionally validate coverage using **split conformal prediction** (Angelopoulos and Bates, 2023), which guarantees marginal coverage of exactly $1 - \alpha = 95\%$ on held-out calibration data.

[PLACEHOLDER] Insert empirical coverage table: for $\alpha \in \{0.90, 0.95, 0.99\}$, report nominal coverage, actual coverage on the test set, and average interval width in basis points. Output of `notebooks/07_uncertainty.ipynb`.

7.2 One-Month Forecast Interval

The 95% forecast interval for the resale price is derived directly from the Monte Carlo simulation in Section 6:

$$\text{CI}_{95\%} = \left[\hat{P}_{0.025}, \hat{P}_{0.975} \right], \quad (11)$$

where $\hat{P}_\alpha$ is the α -quantile of the empirical distribution of $P(t + \Delta t)$ across the $N = 10,000$ simulated paths.

[PLACEHOLDER] Insert one-month forecast confidence interval: a fan chart showing the median forecast and the 80%/95% prediction bands over the month ahead, for a representative sample loan. Output of `notebooks/06_monte_carlo_forecast.ipynb`, Figure 6.

8 Conclusion

We have presented an end-to-end quantitative framework for pricing term loan interest rates. The key contributions are:

1. A **two-stage ML model** (PD classifier + spread regressor) that achieves strong out-of-sample accuracy on historical syndicated loan data.
2. An extension to **private and unlisted borrowers** via the Altman Z"-Score, an IRB scorecard, and transfer learning from public firm financial statements.
3. A **one-month resale price forecast** based on a calibrated Ornstein-Uhlenbeck spread dynamics model and Monte Carlo simulation.
4. **Calibrated 95% confidence intervals** via quantile gradient boosting, validated by split conformal prediction.

[PLACEHOLDER] Insert final model performance summary table: test RMSE, MAE, R^2 , coverage of 95% CI, and mean CI width in bps, compared against a benchmark (e.g., OLS with Altman Z-Score). Output of `notebooks/08_summary.ipynb`.

A Mathematical Derivations

A.1 Merton (1974) Bond Pricing Formula

Under the Merton framework, the value of a risky zero-coupon bond with face value F and maturity T is:

$$D_0 = Fe^{-rT} - (Fe^{-rT}\Phi(-d_2) - V_0\Phi(-d_1)), \quad (12)$$

where $d_{1,2} = \frac{\ln(V_0/F) + (r \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$ and Φ is the standard normal CDF. The credit spread is $s = -\frac{1}{T} \ln(D_0/Fe^{-rT})$.

A.2 Ornstein-Uhlenbeck MLE Estimation

Given discrete observations $\{s_{t_i}\}_{i=0}^n$ with spacing Δ , the conditional log-likelihood is:

$$\ell(\kappa, \theta, \sigma) = -\frac{n}{2} \ln(2\pi\sigma_\Delta^2) - \frac{1}{2\sigma_\Delta^2} \sum_{i=1}^n (s_{t_i} - \mu_\Delta(s_{t_{i-1}}))^2, \quad (13)$$

where $\mu_\Delta(s) = \theta + (s - \theta)e^{-\kappa\Delta}$ and $\sigma_\Delta^2 = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa\Delta})$.

B Repository Structure

```
loan-pricing/
  data/
    raw/          # Downloaded from FRED, SEC EDGAR, SBA
    processed/    # Cleaned and feature-engineered datasets
  notebooks/
    01_eda.ipynb
    02_pd_model.ipynb
    03_spread_model.ipynb
    04_hyperparameter_tuning.ipynb
    05_private_borrower_example.ipynb
    06_monte_carlo_forecast.ipynb
    07_uncertainty.ipynb
    08_summary.ipynb
  src/
    data/
      fetch_fred.py
      fetch_sec.py
      preprocess.py
    features/
      engineering.py
    models/
      pd_model.py
      spread_model.py
      quantile_model.py
```

```
ou_calibration.py
tests/
requirements.txt
README.md
```

B References

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